SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I Athin S.N.S *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Understanding	20%	2	2	2	2	2
Method Selection	20%	2	2	2	2	2
Solution Process	25%	1	1	1	2	1
Accuracy of Calculation	20%	1	1	1	1	1
Interpretation of Results	15%	1	1	1	1	1

Total Marks 37 **/40**

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

Answer:

Lab A (/26, 255.255.255.192): Network 192.168.20.0 | Hosts 192.168.20.1 – 192.168.20.62 | Broadcast 192.168.20.63 | Usable hosts = 2^6 – 2 = 62

Lab B (/27, 255.255.255.224): Network 192.168.20.64 | Hosts 192.168.20.65 – 192.168.20.94 | Broadcast 192.168.20.95 | Usable hosts = 2^5 – 2 = 30

Lab C (/28, 255.255.255.240): Network 192.168.20.96 | Hosts 192.168.20.97 – 192.168.20.110 | Broadcast 192.168.20.111 | Usable hosts = 2^4 – 2 = 14

Subnets are allocated one after another (block sizes 64, 32, 16) with no overlap.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

Answer:

Block size: 256 – 192 = 64, so subnets start at 0, 64, 128, 192.

System 1 (192.168.1.130): Subnet 192.168.1.128/26 | Range 192.168.1.128 – 192.168.1.191 | Network 192.168.1.128 | Broadcast 192.168.1.191 | Hosts 192.168.1.129 – 192.168.1.190 | Usable = 62

System 2 (192.168.1.190): Same subnet 192.168.1.128/26 | Network 192.168.1.128 | Broadcast 192.168.1.191 | Hosts 192.168.1.129 – 192.168.1.190 | Usable = 62

Result: Both IPs lie within 128 – 191, so each is a valid host in its subnet (and both are in the same subnet). 192.168.1.190 is the last usable address.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

Answer:

HR (/25): Network 172.16.0.0 | Hosts 172.16.0.1 – 172.16.0.126 | Broadcast 172.16.0.127 | Usable = 126

Sales (/26): Network 172.16.0.128 | Hosts 172.16.0.129 – 172.16.0.190 | Broadcast 172.16.0.191 | Usable = 62

IT (/27): Network 172.16.0.192 | Hosts 172.16.0.193 – 172.16.0.222 | Broadcast 172.16.0.223 | Usable = 30

Admin (/28): Network 172.16.0.224 | Hosts 172.16.0.225 – 172.16.0.238 | Broadcast 172.16.0.239 | Usable = 14

Unused range: 172.16.0.240 – 172.16.255.255

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address

for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

Answer:

Subnets created: 2^3 = 8 subnets (/27 borrows 3 bits from the host part).
Usable hosts per subnet: 2^5 – 2 = 30 (block size 32).
172.16.10.78 belongs to: Subnet 172.16.10.64/27 (range 64 – 95).
Network / Broadcast: Network 172.16.10.64 | Broadcast 172.16.10.95 | Hosts 172.16.10.65 – 172.16.10.94
172.16.10.95: It is inside the same range (64 – 95) but is the broadcast address, so it cannot be assigned to a host.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Answer:

Compressed form: 2001:db8::ff00:42:8329 (leading zeros dropped; the run of three 0000 groups replaced by ::)
Expanded form: 2001:0db8:0000:0000:0000:ff00:0042:8329
Network prefix (/64): 2001:0db8:0000:0000::/64, i.e. 2001:db8::/64 (first 64 bits)
Total host addresses: 2^(128 – 64) = 2^64 = 18,446,744,073,709,551,616

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation